

Patient-Aware Contrastive Learning for Paroxysmal Atrial Fibrillation Detection Using RR Intervals

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Abstract—Detecting paroxysmal atrial fibrillation (PAF) from RR-interval (RRI) sequences while preserving each patient’s rhythm characteristics remains a challenging problem. We propose a patient-aware contrastive learning framework whose positive pairs are restricted to segments from the same patient and the same class, paired with a compact multi-branch convolutional encoder and temporal attention pooling. On the IRIDIA-AF dataset, the proposed method attains an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.991. This is significantly higher than supervised contrastive and binary cross-entropy baselines and shows substantially lower variance across random seeds, although threshold metrics such as accuracy are comparable across losses. Without any fine-tuning, the encoder further transfers zero-shot to an SHDB-AF dataset, reaching an AUROC of 0.955. These results show that patient-aware constraints in contrastive learning improve AF detection and highlight the value of preserving inter-patient variability for reliable real-world deployment.

Index Terms—atrial fibrillation, contrastive learning, RR intervals, wearable monitoring, patient-aware representation learning, paroxysmal AF detection

I. INTRODUCTION

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia, affecting over 37 million people worldwide and carrying a fivefold elevated risk of ischaemic stroke [1]. PAF, in which episodes self-terminate within less than seven days, represents the earliest clinically actionable stage. Reliable detection opens a window for pharmacological and lifestyle interventions before progression to permanent forms [2]. Yet PAF is episodic and often asymptomatic, making incidental detection during routine clinic visits unreliable. Continuous, long-term ambulatory monitoring is therefore essential.

Modern smartwatches and single-lead wearable patches detect cardiac R-peaks natively and stream RR intervals (RRI) sequences at negligible power cost, requiring no full electrocardiogram (ECG) waveform storage or transmission. RRIs are sensitive to the autonomic and electrophysiological changes

that precede AF onset [3], [4]. Because RRI streams are compact, small models can run directly on-device, enabling private real-time screening. In resource constrained settings where clinical grade 12-lead ECG monitoring is scarce, this offers a realistic path to early AF detection at scale.

Despite strong population level results, supervised deep learning models trained on RRI dynamics often generalize poorly to patients whose normal baseline differs substantially from the training population [3]. This limitation is particularly severe for PAF, where episode characteristics vary to a great extent across individuals. Roughly one third of monitored patients remain asymptomatic [5]. Existing contrastive approaches for cardiac signals treat all same class samples as interchangeable positives [6]–[10]. This collapses the rich inter-patient structure of sinus rhythm (SR) and AF dynamics into a single shared prototype per class. Patient-level contrastive learning has shown value in the broader ECG domain [11], [12]. We build on this line of work and adapt it to AF detection from RRI sequences through a same-patient same-class positive-pair construction tailored to that setting.

To address this gap, we propose a patient-aware contrastive learning framework for PAF detection from RRI sequences. The key contributions of this work are: (i) a patient-aware contrastive loss that restricts positive pairs to same-patient same-class segments, explicitly preserving each patient’s individual rhythm manifold; and (ii) a compact multi-branch convolutional encoder with temporal attention pooling, evaluated via frozen encoder linear probing on a strictly patient-independent test split.

II. RELATED WORK

Early automated PAF detection relied on handcrafted heart rate variability (HRV) features extracted from RR intervals, including time-domain statistics, frequency-domain power spectral density, and nonlinear measures such as sample entropy and the Poincaré plot [13]–[16]. Udawat et al. reached 94.43%

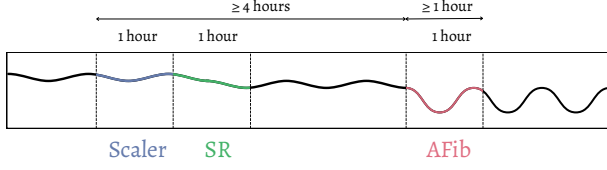


Fig. 1. Illustration of the episode selection and preprocessing pipeline. SR normalization windows (hour 0–1) are disjoint from classification windows (hours 1–2), ensuring the scaler has no access to labeled data.

accuracy on the MIT-BIH AF database [17] with a Fourier Decomposition Method [13]. While interpretable, such methods require careful feature engineering and do not naturally accommodate the inter-patient rhythm variability that characterizes PAF.

Recurrent neural network (RNN) and convolutional neural network (CNN) architectures have largely superseded hand-crafted features for AF classification from RR intervals. Faust et al. report 98.51% accuracy with a long short-term memory (LSTM) network on 100-beat sliding windows under 10-fold cross-validation [18], and Andersen et al. a combined CNN-RNN model with 98.98% sensitivity under five-fold cross-validation [19]. On records from patients held out from training, however, the same authors report that their sensitivity falls to 86.04%, over 12 points lower. Such a gap shows that strong cross-validation performance need not generalize across individuals, and it motivates the present work.

Self-supervised and contrastive methods have recently shown promise for ECG representation learning [20]–[22]. Supervised contrastive learning [10] extends instance level objectives to incorporate class labels, pulling all same-class embeddings together while repelling different class pairs. Applied to cardiac signals, this formulation treats SR segments from all patients as interchangeable, discarding patient-specific differences and degrading AF detection accuracy, as described in Section III-E.

III. METHODOLOGY

A. Datasets

The IRIDIA-AF dataset [23] was used in this work, comprising long-term single-lead ECG recordings from 167 patients with paroxysmal AF. An episode was considered valid if the AF duration was ≥ 1 hr, the immediately preceding SR duration was ≥ 4 hr, and all beats resided within a single continuous file. These two thresholds are deliberate design choices rather than established clinical criteria. The ≥ 4 hr preceding-SR requirement enforces a temporal margin between the SR segments used for classification and AF onset. Because autonomic and heart rate variability changes are known to emerge in the minutes to hours before an episode [3], [4], SR windows drawn immediately before onset would capture pre-AF transitional rhythm rather than clean sinus rhythm and contaminate the negative class. The ≥ 1 hr AF requirement ensures each episode is a sustained, unambiguous

segment that yields sufficient windows for reliable labeling. Together these criteria trade a smaller number of eligible episodes for cleaner, better-separated SR and AF classes. Splits were constructed at the patient level to prevent data leakage: 119, 24, and 24 patients were allocated to training, validation, and testing, respectively. After episode-quality filtering, 154, 27, and 24 episodes were retained across the three splits, with comparable episode counts in each.

For cross-dataset evaluation we additionally use the SHDB-AF database [24], a public Japanese Holter ECG corpus of atrial fibrillation. It is held out entirely from training and used only for the zero-shot test in Section IV-D. After identical preprocessing and episode-quality filtering, 2980 windows from 16 patients were retained (1739 AF, 1241 SR). The MIT-BIH AF database (AFDB) [17] was unsuitable for this purpose, as no recording satisfies our episode-quality criteria (AF ≥ 1 hr and preceding SR ≥ 4 hr).

B. Data Preprocessing

A robust scaler normalized the RRI sequences (RR_i) to remove each patient’s baseline heart rate. Here $\mathcal{R}_{\text{scale}}^{(p)}$ is the RRI sequence for patient p and $\tilde{RR}_i$ is the normalized RRI sequence, and $\text{IQR}(\cdot)$ denotes the interquartile range.

$$\tilde{RR}_i = \frac{RR_i - \text{median}(\mathcal{R}_{\text{scale}}^{(p)})}{\text{IQR}(\mathcal{R}_{\text{scale}}^{(p)})}. \quad (1)$$

The first hour of SR was used to fit the robust scaler, the most temporally stable portion of the recording. SR windows for classification task were drawn from the subsequent hour (hours 1–2), which is strictly disjoint from the normalization fit (Figure 1). The ≥ 4 hr SR inclusion criterion further ensures that all SR windows begin at least 2 hr before AF onset, reducing the risk of including pre-episode transitional rhythms. Each normalized RRI sequence was segmented using a sliding window of $W = 200$ beats with a stride of $S = 50$ beats, yielding windows labeled SR (0) or AF (1). Windows containing any beat shorter than 200 ms or longer than 2000 ms were excluded as physiologically implausible. The values of W and S were selected by optimizing hyperparameters, as described in Section III-G.

C. Patient-Aware Mini-Batch Sampling

Each mini-batch was constructed by sampling from P patients, with each patient contributing n SR and n AF windows, yielding a batch size of $B = P \times 2n$. A patient was eligible for a given batch only if both its SR and AF pools contained at least n windows. This guarantees that every batch contains same-patient same-class pairs for positive comparisons, enforces strict class balance across patients, and prevents high-episode patients from dominating the training signal.

D. Encoder Architecture

The proposed encoder is kept compact while retaining the capacity to capture multi-scale RRI dynamics. It consists of

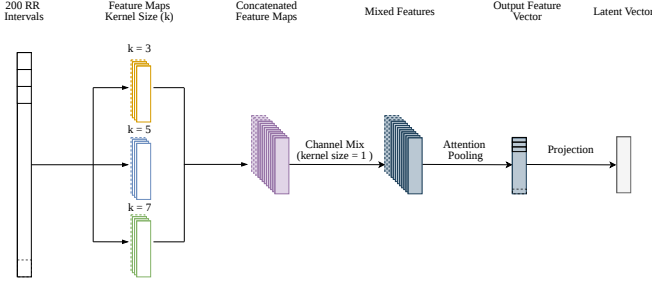


Fig. 2. Encoder architecture: a multi-branch convolutional backbone with temporal attention pooling followed by a two-layer Multi-Layer Perceptron (MLP) projection head.

three components: a multi-scale CNN backbone, a temporal attention pooling module, and a projection head.

1) *Multi-Scale CNN Backbone*: Three parallel 1D CNN branches with kernel sizes $k \in \{3, 5, 7\}$ capture beat-to-beat fluctuations, medium range oscillations, and broader trend dynamics simultaneously, motivated by Inception style multi-scale processing [25]. Each branch applies three strided convolution blocks (stride = 2) with channel depths of 16, 32, and 64, respectively. Group normalization [26] with 8 groups and rectified linear unit (ReLU) activations are applied after each convolution.

2) *Temporal Attention Pooling*: Branch outputs are concatenated and fused via a pointwise (1×1) convolution that performs cross branch feature mixing. Rather than applying global average pooling, a temporal attention module scores each time step independently [27],

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \mathbf{h}_t)}{\sum_{t'} \exp(\mathbf{w}^\top \mathbf{h}_{t'})}, \quad \mathbf{z} = \sum_t \alpha_t \cdot \mathbf{h}_t. \quad (2)$$

This formulation allows the encoder to up-weight physiologically salient RRI segments without requiring manual annotation, and is intended to help with AF episodes that have variable onset dynamics.

3) *Projection Head*: Following SimCLR [9], a two-layer MLP maps the attention pooled latent representation $\mathbf{z}$ to a lower dimensional embedding $\mathbf{e}$. Both $\mathbf{e}$ and $\mathbf{z}$ are ℓ_2 -normalized to the unit hypersphere, making cosine similarity and dot-product equivalent and stabilizing the contrastive loss landscape [28]. The contrastive objective operates on the projected embedding $\hat{\mathbf{e}}$; downstream linear evaluation uses the encoder output $\hat{\mathbf{z}}$.

E. Patient-Aware Contrastive Objective

For an anchor i with class label y_i and patient identifier p_i , the positive and negative sets are defined as,

$$\mathcal{P}(i) = \{j \neq i \mid p_j = p_i \wedge y_j = y_i\}, \quad (3)$$

$$\mathcal{N}(i) = \{j \neq i \mid y_j \neq y_i\}. \quad (4)$$

Here $\mathcal{P}(i)$ is the set of positive pairs and $\mathcal{N}(i)$ is the set of negative pairs. Critically, SR segments from different patients

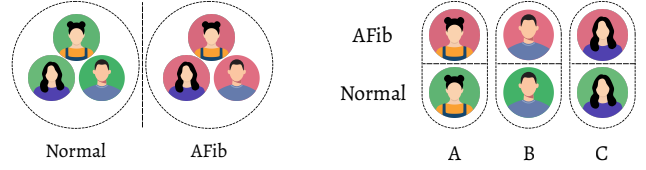


Fig. 3. Comparison of standard supervised contrastive learning (left), which treats all same class segments as positives regardless of patient origin, and the proposed patient-aware formulation (right), which restricts positive pairs to same-patient same-class segments.

are excluded from $\mathcal{P}(i)$, which prevents the encoder from collapsing distinct individual SR baselines into a single shared prototype. This is the key departure from standard supervised contrastive learning [10], as illustrated in Figure 3. The per-anchor contrastive loss is,

$$\mathcal{L}_i = -\log \frac{\sum_{j \in \mathcal{P}(i)} \exp(s_{ij})}{\sum_{j \in \mathcal{P}(i)} \exp(s_{ij}) + \sum_{k \in \mathcal{N}(i)} \exp(s_{ik})}, \quad (5)$$

where $s_{ij} = \hat{\mathbf{z}}_i^\top \hat{\mathbf{z}}_j / \tau$ and τ is a learnable temperature parameter. The batch loss is the mean over all anchors: $\mathcal{L} = \frac{1}{B} \sum_i \mathcal{L}_i$.

F. Training Procedure

The encoder was implemented in PyTorch and trained on a single NVIDIA RTX 2080 GPU. The latent and projection dimensions were 128, with 0.12 dropout on the projection head. We used AdamW (learning rate 6.8×10^{-3} , weight decay 8.8×10^{-4}) with cosine annealing to a minimum of 10^{-6} , and a learnable temperature τ initialized to 0.05. Patient aware sampling used $P = 8$ patients and $n = 16$ windows per class per patient ($B = 128$). Training ran for up to 100 epochs with early stopping on validation AUROC (patience 10).

G. Hyperparameter Selection

Hyperparameters were selected via Bayesian optimization using the Optuna Tree-structured Parzen Estimator (TPE) sampler [29], with validation AUROC as the objective. The 32 jointly optimized hyperparameters spanned encoder architecture (kernel sizes, channel depths, group-norm groups), training (learning rate, weight decay, dropout, temperature), and data pipeline settings (window size W , stride S). All search used the validation split only, with the test split held out until final evaluation.

IV. EXPERIMENTS AND RESULTS

A. Evaluation Protocol

The standard contrastive learning evaluation protocol was adopted. The trained encoder was frozen, and a logistic regression linear probe was fit on training set embeddings. The probe was then evaluated on the held out test split, which comprises entirely unseen patients. This protocol measures the quality

TABLE I
TEST SET PERFORMANCE OF THE PROPOSED LOSS FUNCTION (4160 AF
AND 3799 SR SEGMENTS FROM UNSEEN PATIENTS; MEAN $\pm$ STD ACROSS
50 MODEL SEEDS). BOLD INDICATES THE PRIMARY EVALUATION
METRICS.

Class	Prec.	Recall	F1
SR (0)	0.981 ± 0.012	0.920 ± 0.020	0.949 ± 0.012
AF (1)	0.931 ± 0.016	0.984 ± 0.010	0.957 ± 0.010
Accuracy		0.953 ± 0.010	
AUROC		0.991 ± 0.005	

of the representation independently of downstream classifier capacity. All experiments were repeated over 50 independent model seeds, controlling weight initialization and mini-batch ordering on a fixed patient split. The results are reported as mean $\pm$ standard deviation across runs.

B. Classification Results

Table I reports test set performance of the proposed loss function on 4160 AF and 3799 SR segments from unseen patients, averaged across seeds. The proposed loss achieves a mean AUROC of 0.991 ± 0.005 and overall accuracy of 0.953 ± 0.010 . AF recall (0.984 ± 0.010) consistently exceeds SR recall (0.920 ± 0.020) across all seeds with a clinically favorable asymmetry which minimizes missed AF detections at the cost of a modest increase in false positives. On average, 4093 of 4160 AF segments and 3494 of 3799 SR segments are correctly classified, demonstrating strong generalization to unseen patients. The relatively higher variance in SR recall ($\sigma = 0.020$) reflects patient level heterogeneity in baseline sinus rhythm rather than model instability, as the corresponding SR precision remains stable (0.981 ± 0.012).

C. Effect of the Patient-Aware Constraint

Table II presents a full comparison of three loss functions: the proposed patient-aware loss, standard supervised contrastive loss (SupCon) [10], and binary cross-entropy (BCE). These were evaluated on the same fixed patient-independent test split. The encoder architecture, training budget, and linear probe evaluation protocol are identical across all three conditions. Any difference in performance is therefore attributable solely to the choice of loss function.

The most notable advantage of the proposed loss is its training stability. Compared with SupCon, it reduces the seed-to-seed AUROC standard deviation by $1.6\times$. Compared with BCE, it reduces the seed-to-seed AUROC standard deviation by $2.5\times$. This reduction is statistically significant (Levene test; Table II). This suggests that the loss landscape is less sensitive to weight initialization, which is important for clinical deployment. The proposed loss also achieves a significantly higher mean AUROC than both baselines, with a medium-to-large effect size (Table II). In contrast, accuracy and F1 are statistically similar across losses. Overall, AUROC and seed stability, rather than threshold-dependent metrics, are the main

dimensions along which the patient-aware constraint provides a reliable advantage.

D. Cross-Dataset Generalization

We evaluate the model trained on IRIDIA-AF zero-shot on the SHDB-AF cohort (Section III-A). The frozen encoder and the linear probe, both fit only on IRIDIA-AF, are applied without any fine-tuning or re-normalization to the new cohort. The model attains an AUROC of 0.955 and 0.88 accuracy, with AF detected at 0.95 recall / 0.85 precision (0.90 F1) and SR at 0.77 recall / 0.91 precision (0.84 F1). AF sensitivity is thus largely retained across a different acquisition device, population, and country. The lower SR recall (0.77) and the AUROC drop from 0.991 to 0.955 indicate a harder target distribution, reflecting shifts in baseline heart rate and recording conditions. Because the baselines were not evaluated on this cohort, the result establishes that the learned representation transfers zero-shot but does not demonstrate an advantage over them.

E. Comparison with Prior Work

Direct numerical comparison with prior work is limited by differences in datasets and evaluation methodology (Table III): cross-validation with within-patient mixing inflates reported metrics, as Andersen et al.’s own sensitivity drop on unseen patients illustrates. Under a strict patient-independent holdout protocol, the proposed method achieves $98.4\% \pm 1.0\%$ sensitivity. Our SR class is also drawn from sinus rhythm at least 2 hr before AF onset, so the negative class reflects clean rather than transitional rhythm. This removes ambiguous pre-onset segments that a deployed screening system would additionally need to handle.

V. DISCUSSION

The consistent AF recall of 0.984 ± 0.010 across random seeds is particularly relevant in the clinical context of paroxysmal AF screening. A missed AF episode may delay anticoagulation therapy and leave a patient at elevated stroke risk. High sensitivity is therefore the primary objective. The false positive rate implied by AF precision of 0.931 ± 0.016 is acceptable for a screening system. In such a system, flagged segments would be reviewed by a clinician rather than acted upon autonomously. The patient-independent evaluation protocol is a deliberate, conservative choice that reflects real-world deployment more faithfully than cross-validation.

Standard supervised contrastive learning imposes a single shared SR prototype by pulling together SR segments from all patients, a fundamental limitation for paroxysmal AF where healthy SR baselines differ substantially across individuals due to age, fitness, medication, and autonomic tone. The patient-aware constraint instead lets the encoder learn per-patient SR manifolds while preserving global SR versus AF separability. Figure 4 shows this qualitatively: same-patient segments remain grouped rather than collapsing into a single SR prototype, consistent with the constraint preserving each patient’s individual SR baseline. This better-conditioned geometry coincides with the lower seed-to-seed AUROC variance

TABLE II

COMPARISON OF LOSS FUNCTIONS ON A FIXED PATIENT-INDEPENDENT TEST SPLIT ACROSS 50 INDEPENDENT MODEL SEEDS (MEAN $\pm$ STD). FOR EACH BASELINE, COHEN’S d AND THE PAIRED t -TEST p QUANTIFY ITS AUROC DIFFERENCE FROM THE PROPOSED LOSS, AND LEVENE p TESTS THE DIFFERENCE IN SEED VARIANCE; A WILCOXON SIGNED-RANK TEST AGREES WITH THE t -TEST. BOLD INDICATES THE BEST MEAN PER METRIC. AUROC AND SEED VARIANCE ARE THE PRIMARY DIFFERENTIATING METRICS; ACCURACY AND F1 DIFFERENCES ARE NOT STATISTICALLY SIGNIFICANT.

Loss Function	AUROC	AF Recall	AF F1	Accuracy	d	t -test p	Levene p
Proposed Loss	0.991 $\pm$ 0.005	0.984 $\pm$ 0.010	0.957 $\pm$ 0.010	0.953 $\pm$ 0.010	—	—	—
SupCon [10]	0.986 $\pm$ 0.007	0.972 $\pm$ 0.009	0.954 $\pm$ 0.010	0.951 $\pm$ 0.011	0.73	4.8×10^{-6}	7×10^{-3}
BCE	0.980 $\pm$ 0.012	0.975 $\pm$ 0.009	0.956 $\pm$ 0.011	0.953 $\pm$ 0.012	0.95	2.1×10^{-8}	5×10^{-6}

TABLE III

PERFORMANCE COMPARISON WITH RELATED WORK ON RR INTERVAL BASED AF DETECTION. DIRECT NUMERICAL COMPARISON IS LIMITED BY DATASET AND EVALUATION DIFFERENCES.

Study	Method	Dataset	Validation	Sensitivity	Specificity	Accuracy
Udawat et al. [13]	HRV + ML	MIT-BIH AF	Cross validation [†]	95.16%	92.46%	94.43%
Andersen et al. [19]	CNN + RNN	MIT-BIH AF + 2 others	5-fold CV [†]	98.98%	96.95%	—
Andersen et al. [19] (holdout)	CNN + RNN	Unseen records	Patient holdout	86.04%	98.96%	—
Faust et al. [18]	LSTM	MIT-BIH AF	10-fold CV [†]	—	—	98.51%
Faust et al. [18] (blindfold)	LSTM	MIT-BIH AF (3 subj.)	3-subject holdout	—	—	99.77%
Ours	CNN + MLP	IRIDIA-AF v1.0.1	Patient holdout	98.40 $\pm$ 1.03%	91.97 $\pm$ 2.03%	95.33 $\pm$ 1.05%

[†] Cross-validation with within-patient data mixing inflates reported metrics

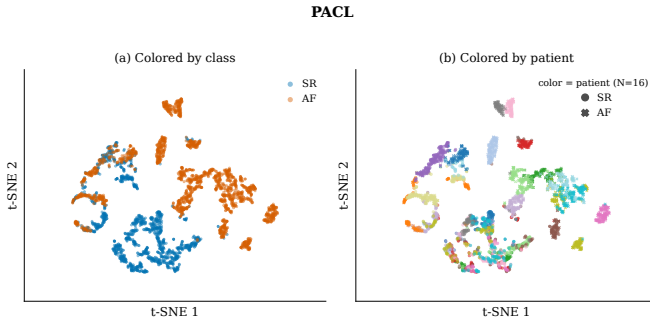


Fig. 4. t-SNE projection of frozen test-set encoder embeddings under the proposed patient-aware objective, colored by class (left) and by patient (right; marker shape denotes class). Beyond the global SR/AF split, segments retain visible per-patient structure, indicating that the encoder preserves individual rhythm characteristics rather than collapsing each class into a single shared prototype.

reported above, indicating a training signal less sensitive to initialization.

A. Limitations and Future Work

This study has several limitations. First, although we show zero-shot transfer to an external dataset (SHDB-AF), the evaluation is still limited to two Holter ECG cohorts and a single patient split. As a result, the reported seed variance mainly reflects initialization and batch ordering, not variability across different data splits. Testing on additional populations, more splits, and consumer wearable recordings would strengthen the evidence. Second, the linear probe measures representation quality but is not a full end-to-end deployment. Fine-tuning or patient-level adaptation may yield further gains.

Several directions are promising for future work. A natural next step is to extend the patient-aware framework to a few-shot setting, where a few labeled segments from a new patient are used to personalize the linear probe for wearable deployment. Another useful direction is a systematic ablation study to isolate the contribution of each architectural component, such as temporal attention pooling and the multi-branch backbone. Prospective validation on a diverse multi-centre cohort would be an important step before clinical translation.

VI. CONCLUSION

We proposed a patient-aware contrastive learning framework for PAF detection from RRI sequences. By restricting positive pairs to same-patient same-class segments, the proposed loss prevents the encoder from collapsing distinct individual SR baselines into a single shared prototype, which is particularly harmful for PAF where inter-patient rhythm variability is substantial. Evaluated on the IRIDIA-AF dataset under a strict patient-independent protocol, the proposed method achieves an AUROC of 0.991 ± 0.005 , significantly higher than both supervised contrastive and binary cross-entropy baselines ($p < 10^{-5}$), and reduces seed-to-seed AUROC standard deviation by $1.6\times$ and $2.5\times$ respectively, while remaining comparable on accuracy and F1. These results demonstrate the effectiveness of patient identity conditioning as a training signal and highlight the potential of fine-grained contrastive objectives for robust, generalisable PAF screening on wearable devices.

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